Agentic AI

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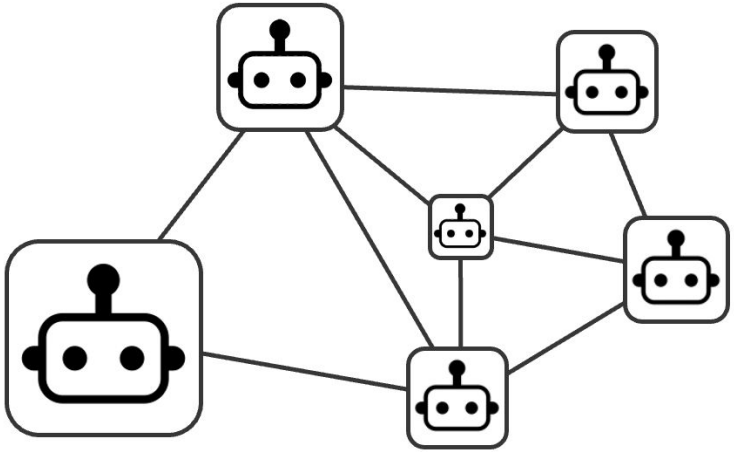
https://abril4416.github.io/



Image

Prior to Pre-training

- One model one task:
  - Reading comprehension: An example from SQuAD

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

Prior to Pre-training

- One model one task:
  - Sentiment classification

"I love this movie.
I've seen it many times
and it's still awesome."

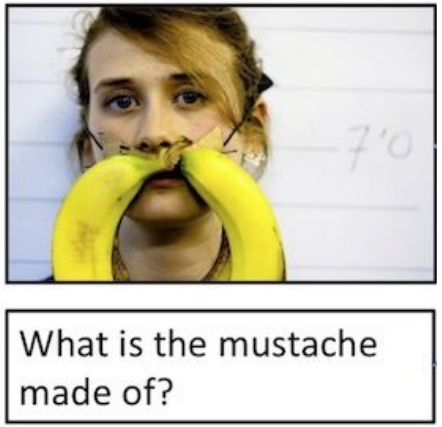
"This movie is bad.
I don't like it it all.
It's terrible."

Prior to Pre-training

- One model one task:
  - Machine translation

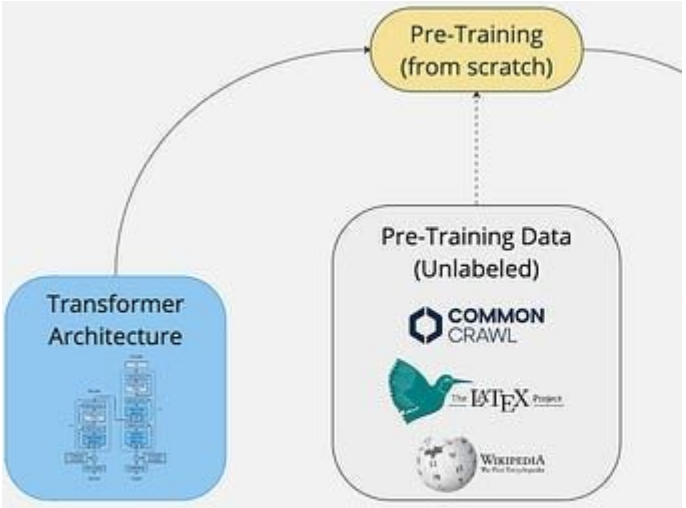
Prior to Pre-training

- One model one task:
  - Image question answering, phrase grounding



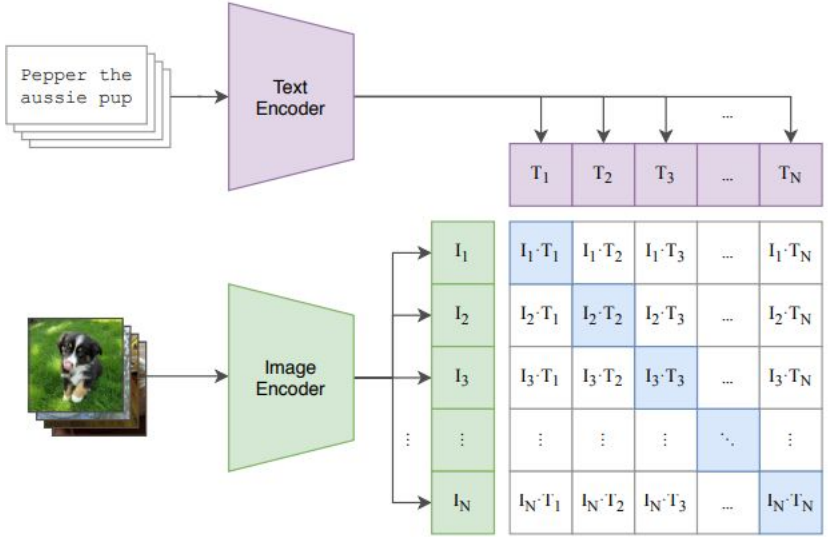
Pre-training

- Pre-trained on large data corpus
  - E.g., BERT >3B tokens



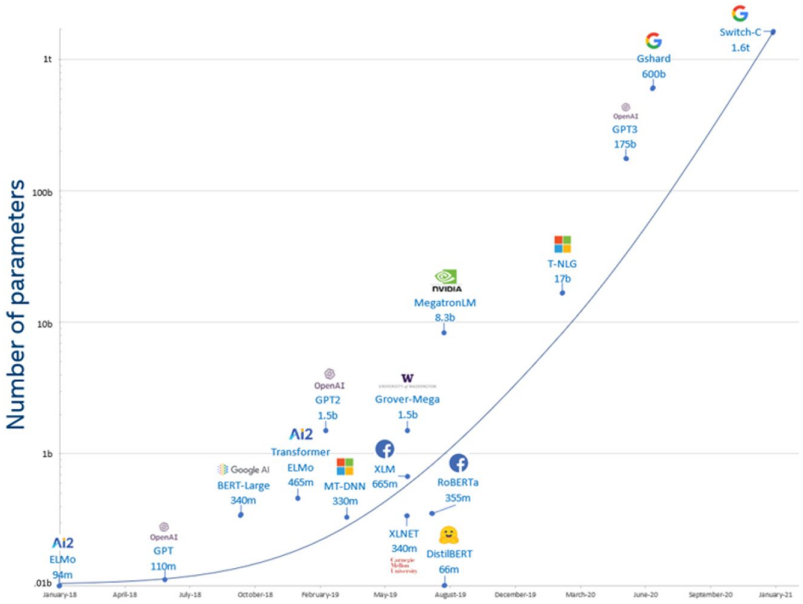
Pre-training

- Pre-trained on large data corpus
  - E.g., CLIP with contrastive learning on 400 millions of IT pairs



Pre-training

- Foundation models
  - Perform well across various tasks (LLMs, MLLMs …)



AI: Are We There Yet?

Real-world Scenario: Complex

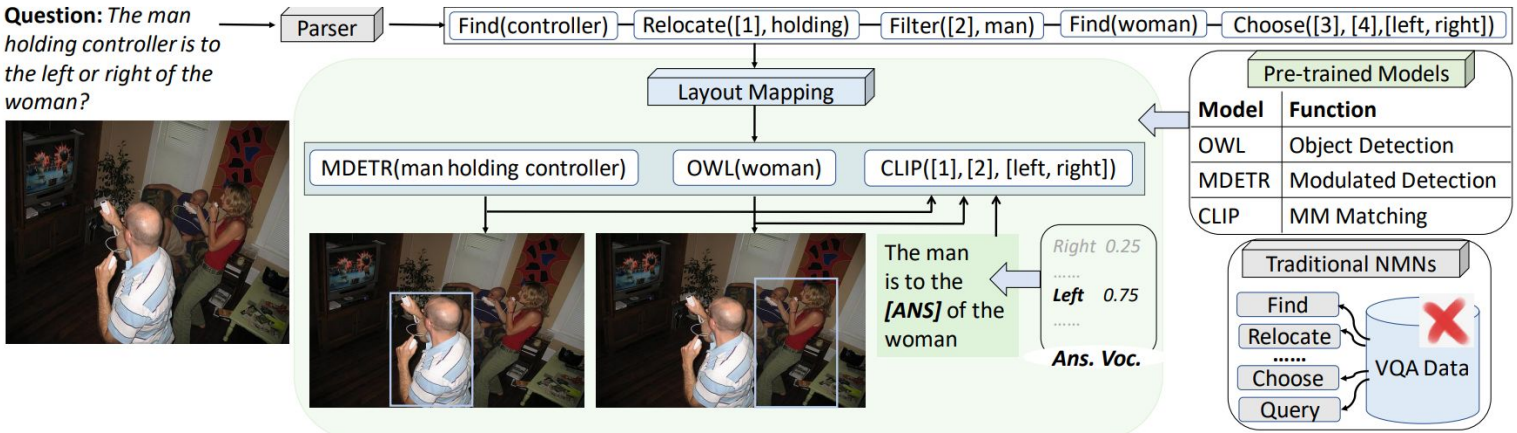
- Multi-step reasoning

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Real-world Scenario: Complex

- Multiple reasoning skills
  - Require the skill of object detection, relational reasoning, spatial reasoning: [paper]



Real-world Scenario: Complex

- Require external knowledge AI models not possess
  - An example from the paper: Black bears are known to inhabit in Yosemite National Park

Question: What California national park are these known to be seen?

Real-world Scenario: Complex

- Two stage of reasoning
  - Acquire relevant knowledge; incorporate knowledge for QA

Question: What California national park are these known to be seen?

Image

Problem with Direct Application

- Weak at multi-step reasoning: wrong answer

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Image

Problem with Direct Application

- Hallucination:
  - E.g., factual fabrication (lack of external knowledge)

Instruction: What are the major environmental impacts of the construction of the Eiffel Tower?

Response: The construction of the Eiffel Tower in 1889 led to the extinction of the Parisian tiger, a species that played a crucial role in the region's ecosystem. Additionally, it is widely recognized as the event that sparked the global green architecture movement.

Image

Problem with Direct Application

- Hallucination:
  - E.g., unfaithful to image (length output; lack of multi-step control)

Generate Detailed Image Description

The image … The first sign displays … "Shakespeare," while … shows "West Beaumont." … In the background, there are two people standing to the left of the sign … A car …

Image

Problem with Direct Application

- Difficult to handle complex tasks requires multiple tools
  - E.g., image-text claim verification (query generation, evidence retrieval, verification, justification generation)

Claim Text: *Kamala Harris with her parents and she is not a black American.*

Claim Image:

Claim Date: *2020.08.12*

Claim Source: *Facebook*

Refuting Reasons: *Misuse of images; Textually refuted*

Image Misuse: *Out-of-Context*

Claim Type: *...*

Q1: *What is the date of the claim image being published?*

Related Image to Q1:

A1: *The image was taken in 2016.*

Q2: *Were Kamala Harris parents alive in 2016?*

A2: *Her mother died in 2009.*

Q3: *Who are the people shown in the Image of the claim?*

Related Image to Q3:

A3: *Harris (center) was with her supporters Suneil Parulekar (left) and Rohini Parulekar (right) at the 2016 Pratham gala.*

Q4: *What is Kamala Harris's ethnic background?*

A4: *She has Jamaican and Indian parents.*

Q5: *Can a person be seen as a black American if they have either Indian or Jamaican parents?*

A5: *Yes. Not all black people are African American.*

Verdict: *Refuted*

Justification: *The image was interpreted out-of-context. The image was taken in 2016, while Harris' mother died in 2009. It is impossible that she was with her parents in the image. Besides, another evidence proves the image shows Harris was with her supporters at the 2016 Pratham gala, rather than with her parents. Harris has Jamaican and Indian heritage, proving she is a black American. Therefore, the textual part of the claim, "she is not a black American" is refuted.*

Summary: Direct Application

- AI models perform well on simple tasks/datasets
- Challenging to handle real-world complex problems
- Leading to false predictions, hallucinations, or an inability to handle

Stand on the Shoulder of Foundation Models

- Leverage foundation models for real-world complex scenarios

Single Step/Model to Pipeline

- Task decomposition:
  - Decompose complex tasks into multiple steps
  - Each step is a simpler task
  - Simplify the task

Single Step/Model to Pipeline

- High-level idea:
  - Input task: → Step 1 → Step 2 → Step 3 → Output
- Each step has a clear role
  - I.e., a sub–task

Single Step/Model to Pipeline

- Example: image-text claim verification
  - Generate information-seeking question
  - Retrieve evidence based on these questions and the claim
  - Verify the claim
  - Generate a justification

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Single Step/Model to Pipeline

- Advantages:
  - Efficient and predictable: smaller models ⇒ larger model performance
  - Easier to debug and evaluate: intermediate results
  - Works well for well-defined tasks
- Limitations:
  - Fail when intermediate errors
  - Limited in general tasks in daily life: not well-defined

Agentic AI: Dynamic Pipeline

- High-level idea: dynamic
  - AI models decided what to do next
  - Not just following recipe

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports',' Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]

Image

Overview

- Foundations of agentic AI
  - Key components and high-level view
  - Design variants of agentic AI components
- Introduction to LangGraph
  - Key components and implementation
  - A simple walkthrough of LangGraph
- AI model for Data Handling
  - Various skills
  - Training recipe for skills
- AI Models for Evaluation

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What is an Agent: Key Components

- A policy operate in an environment
  - Environment: everything outside the agent that it can observe or interact with
- Components:
  - Perception: collects and interprets information from the environment
  - Planning: breaks down complex goals into actionable steps
  - Action: executes decisions by interacting with external systems
  - Memory: stores and retrieves information across time

One Example: Perception

- User query: *Find the best Italian restaurant nearby, check if it's open tonight, and make a reservation for 7pm*

One Example: Planning

- AI action planning: *Find the best Italian restaurant nearby, check if it's open tonight, and make a reservation for 7pm*
  - Search for nearby Italian restaurants
  - Rank/select best option (e.g., ratings, reviews)
  - Check if the restaurant is open at 7pm
  - Check reservation availability
  - Make reservation

Solution: Action

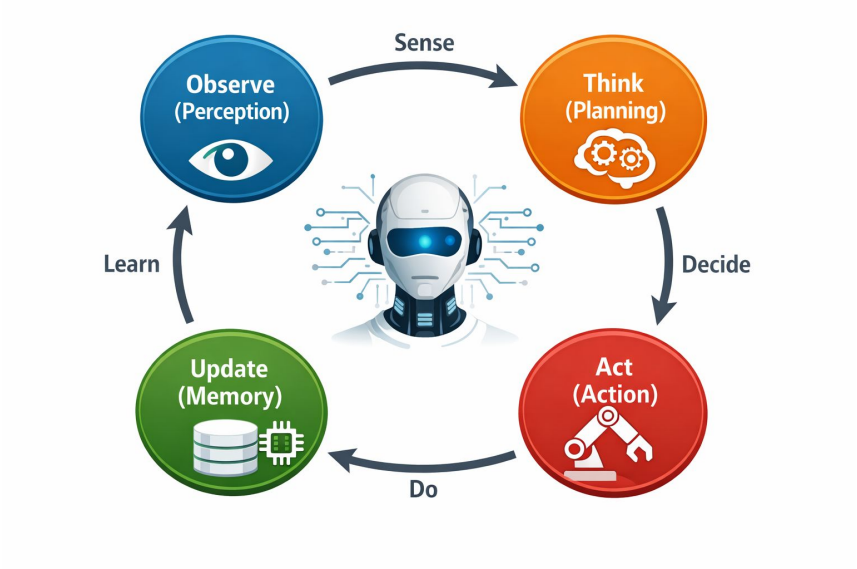
- Step-by-step action execution:
  - Step 1: call Google Maps: returns list of restaurants
  - Step 2: filter by rating: selects top candidate
  - Step 3: open restaurant website:checks opening hours
  - Step 4–5: uses reservation API (e.g., OpenTable); books table at 7pm

One Example: Memory

- Short-term memory: one-time execution
  - List of restaurants retrieved
  - Selected restaurant
  - Reservation details
- Long-term memory: multi-turn summarization
  - User profiles (e.g., food preferences, personality)

Agents: High-level View

- Iteratively
- Loop: Perception → Planning → Action → Memory → …)



Perception

- Interface between environment and agent
- Goal: interpret, extract and normalize inputs
  - Messy message → key information/objective
- Generally, AI models serve as the perception layer
  - LLMs, MLLMs (e.g., images, audio, videos)

Planner

- Goal: decide what to do
  - Determines sequence of actions
  - Chooses tools and strategies
- Planner types
  - Without vs. with feedback

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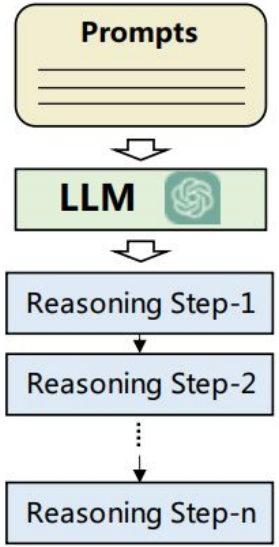
Planner Without Feedback

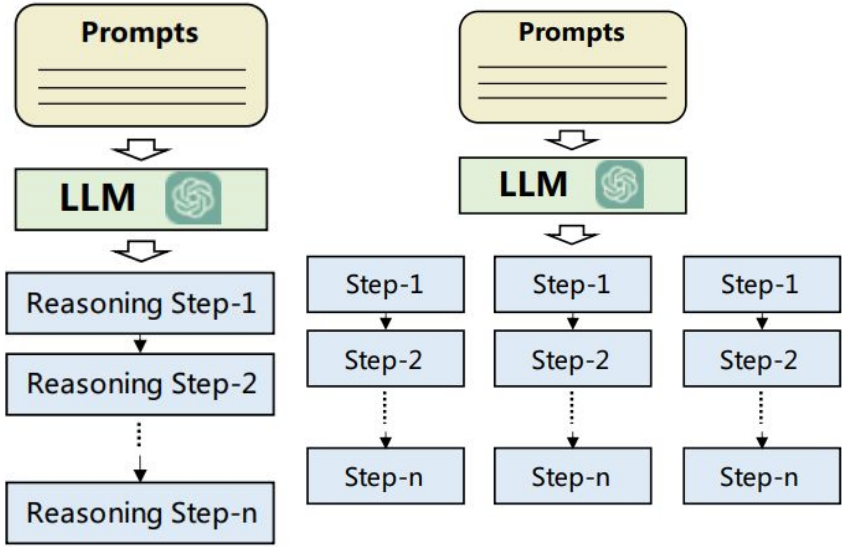
- Do not receive feedback that can influence future planning after taking actions
  - One-time planning and then execution



Image

Planner Without Feedback

- Do not receive feedback that can influence future planning after taking actions
  - Also multi-path reasoning



Image

Planner Without Feedback

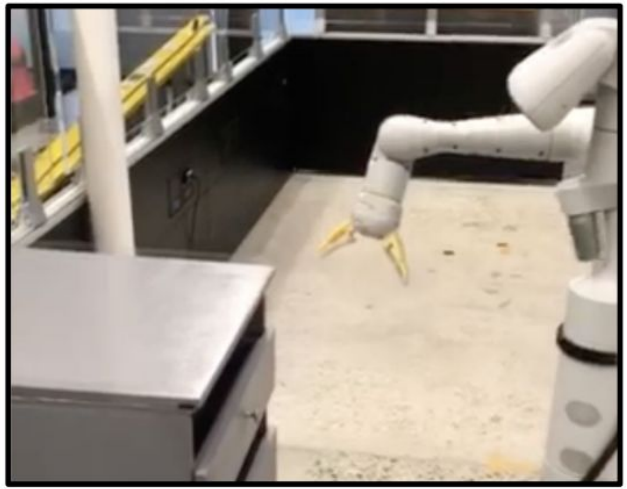
- Problem: less capable towards long-horizon planning tasks
  - Generate an accurate plan initially is hard
  - Inaccurate: initial plan could lead to failures
  - Initial plan non-executable: unpredictable transition dynamics

Planner With Feedback

- Agent can receive feedback after taking action
- Iteratively generate next plan based on observations+feedback
- External feedback:
  - Human
  - Environment
- Internal feedback
  - Agents themselves

Planner With Feedback

- External human feedback: Inner Monologue
  - Specification from human user



Active Scene Description in context:

Robot Action: Go to drawers
Robot Ask: Is the drawer open?
Human: The drawer is closed.
Robot Action: Open the drawer

Planner With Feedback

- External environment feedback
  - Next planning depends on the previous observation
  - Observation is the output from current action

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Image

Planner With Feedback

- Internal feedback
  - if the task fails, it internally provides a critique by suggesting how to complete the task
  - The critique will be used in next planning

Image

Action

- Action: executes decisions from planner
- Directly interacts with the environment
- Types of actions
  - E.g., retrieval, web search, code execution, database query
  - Perception and planning are mostly based on AI models
  - Actions are accomplished with a set of tools

Tool Use for Action

- Some actions AI models are not capable of
  - Online web search, database query, code execution, etc
- Action space: available actions
  - Internal knowledge of AI models
    - Embedded commonsense knowledge
    - Models with specific skills: CodeX, SAM
  - External tools:
    - APIs
    - Query Database/Knowledge base

Learning of Tool Use

- Direct prompting
  - Provide information about executable actions (e.g., definition)
- Supervised fine-tuning
  - Instruction following tool use dataset
    - Teacher model generated: e.g., via Chat-GPT
    - Synthetic data
    - Human annotation

Impact of Actions

- Environment
  - E.g., position moving
- Internal states
  - Memory updates
  - Feedback to future planning: e.g., critique generation
- Trigger new actions
  - E.g., gather sufficient resources, begin construction

Memory

- Type: long / short memory
- Memory storage structure
  - How memory is represented
- Operation on memory
  - What agents can do with memory

Memory Storage

- Vector-based memory
  - Embedding vector (from encoder / LLM embedding model)
- Natural language memory
  - Sequential logs (chat history, observations, actions)
  - Long memory: summarise

Memory Storage

- Vector-based memory:
  - Dense vector embeddings of memory
  - Efficient retrieval at scale (similarity matching)
    - FAISS: Efficient similarity search high-dimensional dense vector
  - Information loss during conversion
    - E.g., structure
  - Hard to edit or update meaningfully

Memory Storage

- Vector-based memory:
- Natural language memory
  - Flexibility: support reasoning, abstraction, reflection …
  - Scale is hard (maximum length of texts)
  - Redundant, verbose
  - Hard for retrieval
  - Hallucination in summarization

Memory Storage

- Vector-based memory:
- Natural language memory
- Symbolic memory
  - Explicit structure, deterministic access
    - E.g., key-value pair, dictionary, tables
  - Precise reasoning: e.g., deduction
  - Compositionality, interpretability
  - Inflexible: schema design; error accumulation in extraction
  - Incapabile: vague concepts, fuzzy queries

Memory Storage

- Vector-based memory:
- Natural language memory
- Symbolic memory: logic
- Graph-based memory: relation
  - More flexible: less rigid than symbolic (pure logic)
    - E.g., vector database
  - Reasoning, interpretability
  - Incapable: implicit relation

Operation on Memory

- Write: add new information to memory
- Read: retrieve information from memory
- Update: change existing memory
- Reasoning: use memory for new insights, decisions and actions

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ANY QUESTION

From Concept to Implementation

- Introduction to LangGraph: stateful, graph-based agent workflows
  - Open-source and free to use
- Key components:
  - Graph:
    - Directed graph: support loops and conditional transitions
    - State: shared memory; all information during execution
    - Nodes: functions that read and update the state
    - Edges: control the flow of execution
  - Tools: a set of APIs, retrieval augmented generator, etc.

Component: State

- Shared memory across steps
- Can design your state
- Input, intermediate results, final output…
- Stored as structured data: dictionary

```
memories[-1].dict()
{'value': {'food_preference': 'I like pizza'},
 'key': '07e0caf4-1631-47b7-b15f-65515d4c1843',
 'namespace': ['1', 'memories'],
 'created_at': '2024-10-02T17:22:31.590602+00:00',
 'updated_at': '2024-10-02T17:22:31.590605+00:00'}
```

Component: State

- Get the state information with *graph.get_state(config)*
  - Return *StateSnapshot* fields
  - Example:

```
StateSnapshot(
    values={'foo': 'b', 'bar': ['a', 'b']},
    next=(),
    config={'configurable': {'thread_id': '1', 'checkpoint_ns': '', 'checkpoint_id':
'1ef663ba-28fe-6528-8002-5a559208592c'}},
    metadata={'source': 'loop', 'writes': {'node_b': {'foo': 'b', 'bar': ['b']}}, 'step': 2},
    created_at='2024-08-29T19:19:38.821749+00:00',
    parent_config={'configurable': {'thread_id': '1', 'checkpoint_ns': '', 'checkpoint_id':
'1ef663ba-28f9-6ec4-8001-31981c2c39f8'}}, tasks=()
)
```

example

Component: Node

- A Python function taking current state and returning updates to it
- START node: send user input to the graph
- END node: terminal node
  - Referenced to denote edges that have no subsequent actions once execution is complete
- Other nodes: python functions

Component: Node

- Add nodes via *graph.add_node(node_name, function)*
  - Allow caching: ttl=3, the cached result is valid for 3 steps

```python
def expensive_node(state: State) -> dict[str, int]:
    # expensive computation
    time.sleep(2)
    return {"result": state["x"] * 2}


builder.add_node("expensive_node", expensive_node, cache_policy=CachePolicy(ttl=3))
```

Component: Edge

- Rules that determine which node runs next: flow of execution
- Fixed edge: always go to next node
  - *graph.add_edge("node_a", "node_b")*
- Conditional edge: depends on state
  - Routing_function accepts current state and returns a value
    - The value is the name of node to sent the state to next
    - Name of the next node
  - *graph.add_conditional_edges("node_a", routing_function)*

Walkthrough

- Example: claim verification (factuality of the claim)
  - E.g., *The Eiffel Tower is in Berlin.*
- Simple pipeline
  - Retrieve evidence
  - Verify claim against evidence
  - Return a verdict

Walkthrough

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Implement with LangGraph

- State
  - Claim, evidence, verdict, justification/reason

```
{
    'claim': 'The Eiffel Tower is in Berlin.',
    'evidence': [],
    'verdict': None,
    'Reason': None
}
```

```
class VerificationState(TypedDict):
    claim: str
    evidence: str
    verdict: str
    reason: str
```

```
from langgraph.graph import StateGraph
graph = StateGraph(VerificationState)
```

Implement with LangGraph

- State
- Node
  - Retrieve_evidence: retrieves and writes evidence into the shared state
  - Not_enough_info: predict not enough information
  - Verify_claim: reads claim and evidence, then writes verdict and reason

Implement with LangGraph

- State
- Node: **omit function details**
  - Retrieve_evidence: retrieves and writes evidence into the shared state
  - Not_enough_info: return Not Enough Information
  - Verify_claim: reads claim and evidence, then writes verdict and reason

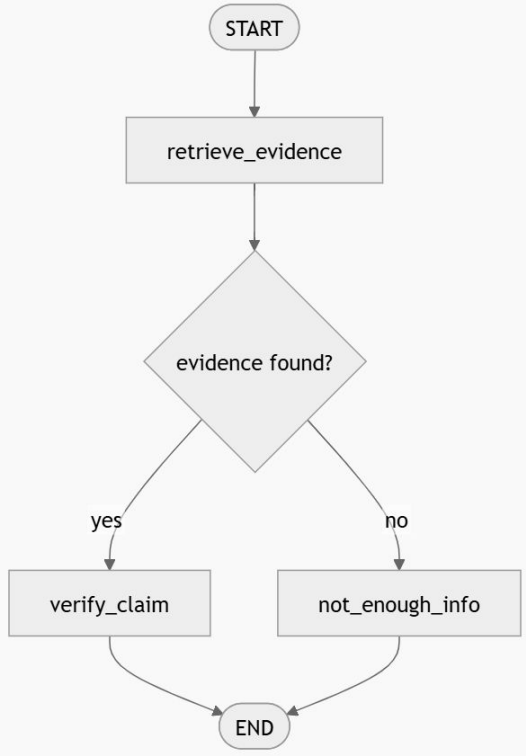
```
graph.add_node("retrieve_evidence", retrieve_evidence)

graph.add_node("verify_claim", verify_claim)

graph.add_node("not_enough_info", not_enough_info)
```

Implement with LangGraph

- State
- Node
- Edge: a simple version
  - In fact, evidence retrieval could be multiple times



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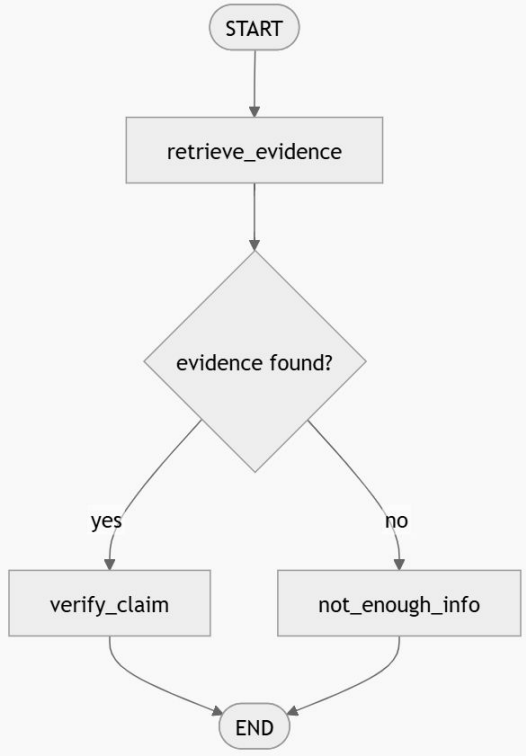


```
graph.add_edge(START, "retrieve_evidence")
graph.add_conditional_edges("retrieve_evidence",
route_after_retrieval)
graph.add_edge("verify_claim", END)
graph.add_edge("not_enough_info", END)
```

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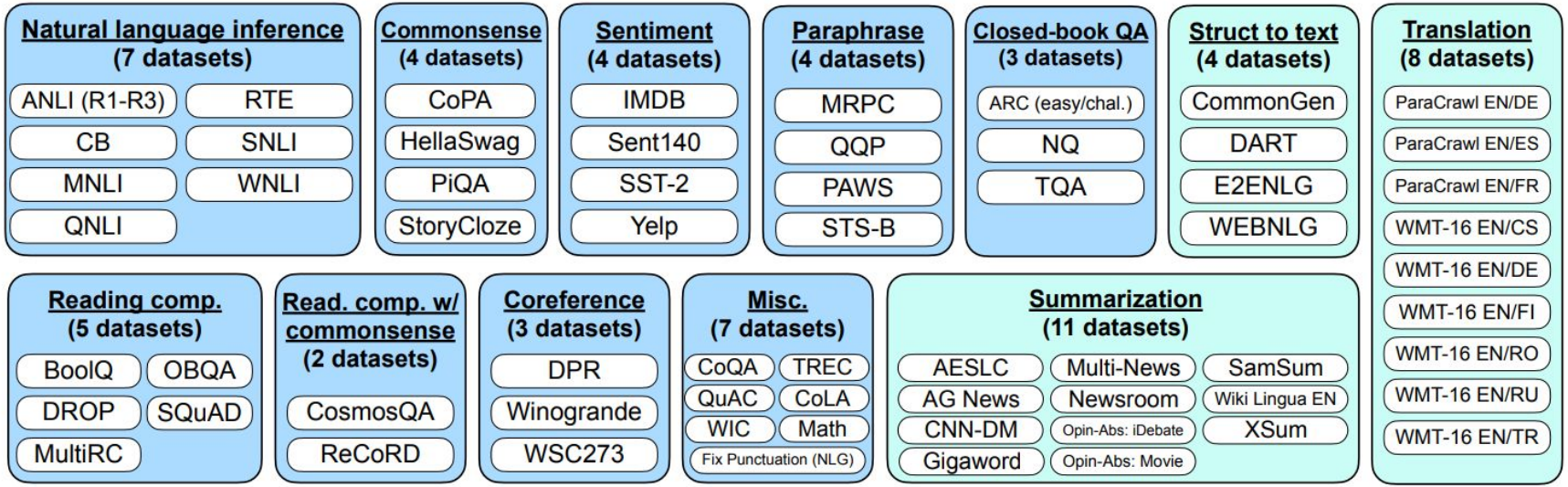
  - Training recipe for skills
- AI Models for Evaluation

ANY QUESTION

AI Models as Tools

- Not limited as chat-bots
- Various skills



example

AI Models as Tools

- Not limited as chat-bots
- Various skills: e.g., date extraction
  - Compare regex vs. LLM based extraction (prompt)
  - Regex: inflexible; find patterns; deterministic
  - LLMs: find meaning

```
Import re
pattern = r"\b(?:\d{1,2}/\d{1,2}/\d{4}|[A-Za-z]+
\d{1,2}, \d{4}|[A-Za-z]+ \d{1,2})\b"

dates = re.findall(pattern, text)
print(dates)
```

Extract all events from the text. Return JSON with:
- event_name
- date (normalized ISO)
- time (if any)
- location (if any)
- description

Skills of AI Models: LLMs

- Parser (e.g., entity extraction)
- Sentiment anlaysis
- Question answering
- Translation
- Summarization
- Coding
- ….

Skills of AI Models: MLLMs

- Object detection
- Reference expression localization
- OCR detection
- Image captioning
- Image question answering
- ….

How to Acquire Skills

- Pre-trained AI models are general-purpose
  - E.g., LLMs can understand language
- Training recipes
  - Further optimized for instruction following, domain specific skills or better reasoner

How to Acquire Skills

- Pre-trained AI models are general-purpose
  - E.g., LLMs can understand language
- Training recipes
- Three core recipes
  - Instruction tuning
  - Continued pre-training
  - Reinforcement learning

Instruction Tuning

- Definition: fine-tune models on (instruction, input → output) pairs
- Goal: Converts models into a task-following assistant

Instruction: Summarize the paragraph
Input: [text]
Output: [summary]

Instruction Tuning

- Definition: fine-tune models on (instruction, input → output) pairs
- Goal: Converts models into a task-following assistant
- Training objective
  - E.g., LLMs, next-token prediction

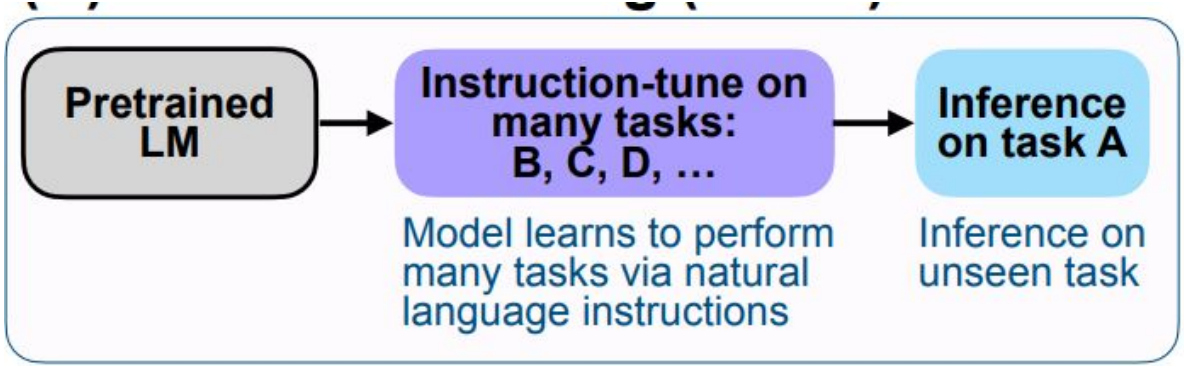
$$\mathcal{L} = -\sum \log P(y \mid x, \text{instruction})$$

Instruction Tuning Dataset

- Synthetic instructions
  - Self-Instruct: 52K instructions
  - Prompt off-the-shelf LM to generate diverse instructions based on seed instruction
- Human written instructions
  - FLAN: multiple paraphrased templates per task
  - Combine 60+ NLP datasets

Instruction Tuning

- Definition: fine-tune models on (instruction, input → output) pairs
- Goal: Converts models into a task-following assistant
- Training objective
- Training outcome:
  - Better instruction following; generalize to unseen tasks



example

Continued Pre-training

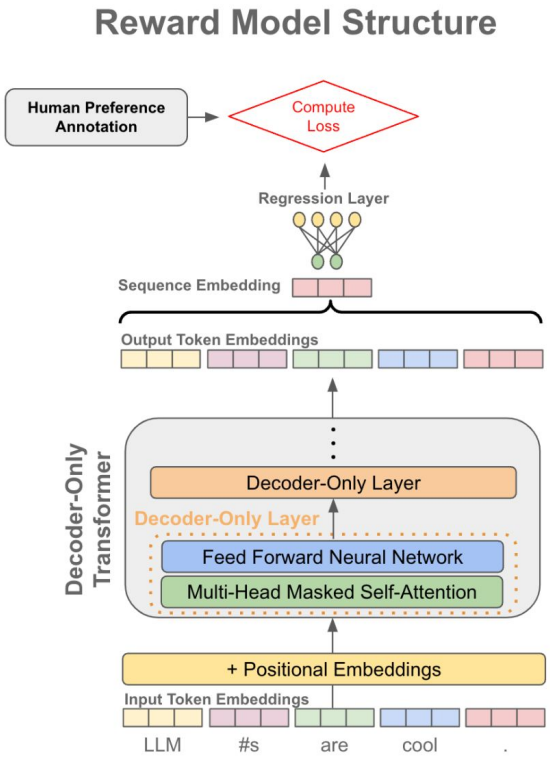
- Further train AI models on raw domain data
  - Code models: CodeLLaMA
    - LLaMA2
    - Trained on Github repositories and StackOverflow
  - Scientific context: ChemDFM
    - LLaMA-13B
    - Trained on 34B tokens from chemical literature and textbooks

Reinforcement Learning

- Tasks require exploration (e.g., multi-step reasoning)
- Comparison with instruction tuning
  - Supervision: ground-truth output vs. reward signal
  - Goal: follow instructions vs. optimize outcomes
  - Advantage: stability vs. exploration

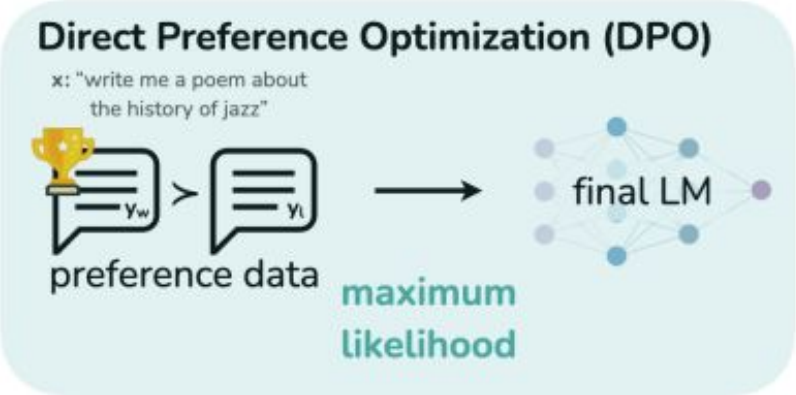
Reinforcement Learning

- RL for safety alignment: LLaMA2-Chat
  - Encourage safer responses
  - Generate responses to maximize rewards
    - Reinforcement learning with human feedback **(RLHF)**
  - Human feedback as reward
    - Safety reward
    - Helpful reward



Reinforcement Learning

- RL for hallucination mitigation
  - Encourage responses with fewer hallucinations
  - Generate responses that are more preferred
    - Direct preference learning **(DPO)**



Reinforcement Learning

- RL for hallucination mitigation
  - Encourage responses with fewer hallucinations
  - Generate responses that are more preferred
    - Direct preference learning **(DPO)**
  - Synthetic data with paired preferred vs. rejected data
    - Higher probability to generate preferred responses
    - Automatic evaluation of hallucination rates, as feedback

Reinforcement Learning

- RL for hallucination mitigation
  - Synthetic data with paired preferred vs. rejected data

Response: The image features two giraffes standing close together in an open area with a mix of grass, dirt, and rocky terrain. They appear to be positioned towards the left side of the scene, possibly as the main subjects. The surroundings include a few trees, one towards the left side and another slightly further to the right. A person is also visible in the image, standing at a distance to the left of the giraffes, possibly observing or taking photographs of the animals.

[Existence]: terrain ✓, giraffe ✓, dirt ✗, person ✓, tree ✓
[Counts]: (two, giraffe) ✓
[Attributes$_{general}$]: (rocky, terrain) ✓, (open, area) ✓, (standing, giraffe) ✓
[Relations$_{general}$]: (giraffe, close to, giraffe) ✓
[Relations$_{spatial}$]: (giraffe, left, scene) ✓, (giraffe, left, image) ✓, (tree, left, scene) ✓, (tree, right, scene) ✗

Reinforcement Learning

- RL for better reasoning: DeepSeekMath
  - Encourage responses with correct answer
  - Which reasoning trajectories are better relative to others for the same problem
    - Better: lead to correct answers
    - Group Relative Policy Optimization **(GRPO)**

$$A_i = r_i - \frac{1}{K}\sum_{j=1}^{K} r_j$$

Reinforcement Learning

- RL for better reasoning: DeepSeekMath
  - Encourage responses with correct answer
  - Which reasoning trajectories are better relative to others for the same problem

$$A_i = r_i - \frac{1}{K}\sum_{j=1}^{K} r_j$$

  - Above-average outputs → positive advantage
  - Below-average outputs → negative advantage

Reinforcement Learning

- RL for better reasoning: DeepSeekMath
  - Encourage responses with correct answer
  - Which reasoning trajectories are better relative to others for the same problem
    - Better: lead to correct answers
    - Group Relative Policy Optimization **(GRPO)**
    - Do not need reward models

Reinforcement Learning

- RL for better reasoning: DeepSeekMath
  - Encourage responses with correct answer
  - Which reasoning trajectories are better relative to others for the same problem
    - Better: lead to correct answers
    - Group Relative Policy Optimization **(GRPO)**
    - Push model towards: 1) better reasoning paths; 2) correct answers

Overview

- Foundations of agentic AI
  - Key components and high-level view
  - Design variants of agentic AI components
- Introduction to LangGraph
  - Key components and implementation
  - A simple walkthrough of LangGraph
- AI model for Data Handling
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ANY QUESTION

Recap: Evaluation

- Necessity for evaluation
  - Evaluation for hallucination

Response: The image features two giraffes standing close together in an open area with a mix of grass, dirt, and rocky terrain. They appear to be positioned towards the left side of the scene, possibly as the main subjects. The surroundings include a few trees, one towards the left side and another slightly further to the right. A person is also visible in the image, standing at a distance to the left of the giraffes, possibly observing or taking photographs of the animals.

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Recap: Evaluation

- Necessity for evaluation
  - Evaluation for evidence from open-web, textual justification

Claim Text: *Kamala Harris with her parents and she is not a black American.*

Claim Image:

Claim Date: *2020.08.12*

Claim Source: *Facebook*

Refuting Reasons: *Misuse of images; Textually refuted*

Image Misuse: *Out-of-Context*

Claim Type: *...*

Q1: *What is the date of the claim image being published?*

Related Image to Q1:

A1: *The image was taken in 2016.*

Q2: *Were Kamala Harris parents alive in 2016?*

A2: *Her mother died in 2009.*

Q3: *Who are the people shown in the Image of the claim?*

Related Image to Q3:

A3: *Harris (center) was with her supporters Suneil Parulekar (left) and Rohini Parulekar (right) at the 2016 Pratham gala.*

Q4: *What is Kamala Harris's ethnic background?*

A4: *She has Jamaican and Indian parents.*

Q5: *Can a person be seen as a black American if they have either Indian or Jamaican parents?*

A5: *Yes. Not all black people are African American.*

Verdict: *Refuted*

Justification: *The image was interpreted out-of-context. The image was taken in 2016, while Harris' mother died in 2009. It is impossible that she was with her parents in the image. Besides, another evidence proves the image shows Harris was with her supporters at the 2016 Pratham gala, rather than with her parents. Harris has Jamaican and Indian heritage, proving she is a black American. Therefore, the textual part of the claim, "she is not a black American" is refuted.*

Recap: Evaluation

- Necessity:
  - Open-ended generation evaluation: one-step in agentic pipeline
    - Retrieved evidence
    - Chat-bot
    - Question answering
    - …
  - Preference data collection
    - Evaluate hallucinations, helpfulness, bias etc.

Recap: Evaluation

- Challenge:
  - Lengthy output, varying semantics, various objectives
- Traditional evaluation metrics
  - Inflexible
  - Incapable towards dynamics in outputs

Recap: Evaluation

- Example in summarization: traditional evaluation metrics fail

GT: The company reported a 20% increase in revenue this quarter, driven primarily by strong growth in its cloud services division, despite a decline in hardware sales.

A: The company's revenue rose 20%, mainly due to cloud services growth, even though hardware sales fell.

B: The company reported revenue and hardware sales changes, including cloud services.

Recap: Evaluation

- Example in summarization: traditional evaluation metrics fail
  - Summary B: bad but overlapping words
  - Summary A: good but with paraphrasing

GT: The company reported a 20% increase in revenue this quarter, driven primarily by strong growth in its cloud services division, despite a decline in hardware sales.

A: The company's revenue rose 20%, mainly due to cloud services growth, even though hardware sales fell.

B: The company reported revenue and hardware sales changes, including cloud services.

Recap: Evaluation

- Example in summarization: traditional evaluation metrics fail
  - Traditional BLEU / ROUGE behavior:
    - B score high: share many surface words (company, revenue, hardware, cloud services …)
    - Reward lexical overlap, not meaning correct

GT: The company reported a 20% increase in revenue this quarter, driven primarily by strong growth in its cloud services division, despite a decline in hardware sales.

A: The company's revenue rose 20%, mainly due to cloud services growth, even though hardware sales fell.

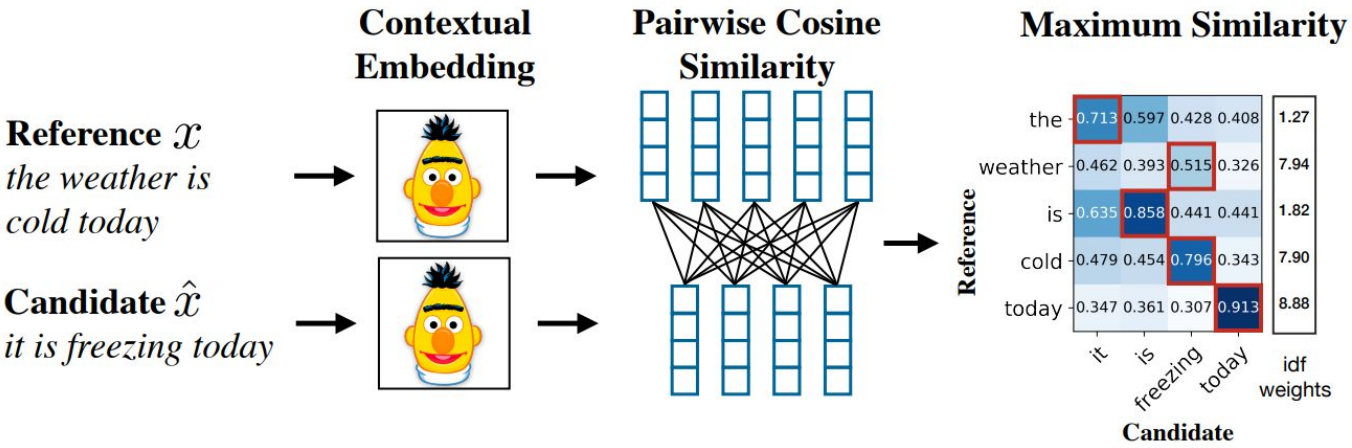
B: The company reported revenue and hardware sales changes, including cloud services.

AI Models as Evaluator

- Intuition: evaluation is easier than generation
- A good evaluator: aligns well with human judgement
  - Pearson/Spearman correlation coefficient
    - Between evaluator scoring and human scoring
  - Fleiss' kappa, free-marginal multi-rater kappa
    - Classification between evaluator and human

AI Models as Evaluator

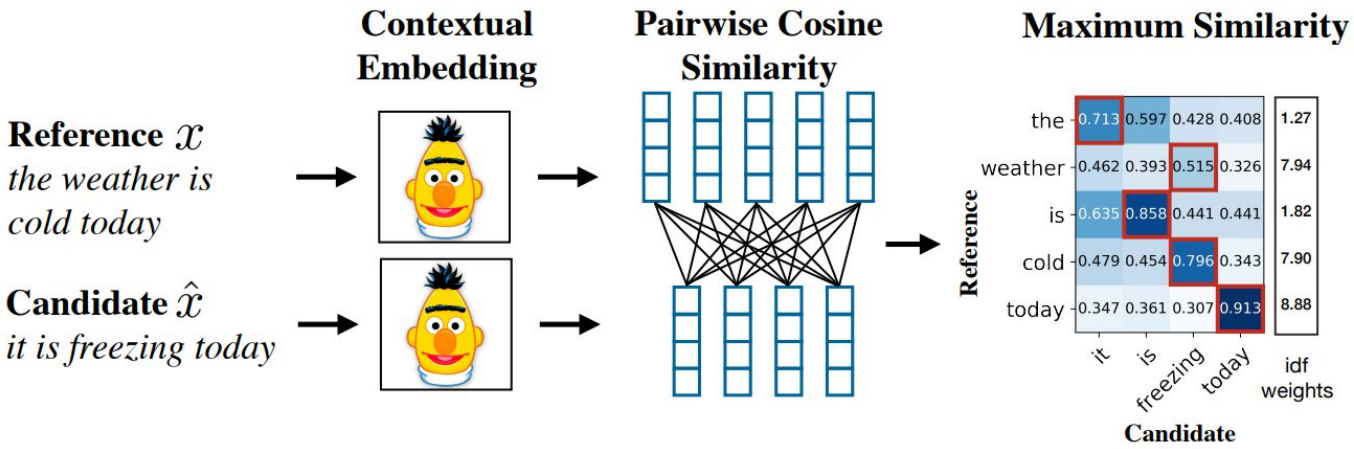
- BERTScore:
  - Pairwise similarity
  - Greedy matching



AI Models as Evaluator

- BERTScore:
  - Pairwise similarity; greedy matching
  - Compute recall and precision

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j \quad , \quad P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$



AI Models as Evaluator

- BERTScore:
  - Pairwise similarity; greedy matching
  - Compute recall and precision
    - How likeyly tokens in the reference could be found in the prediction: recall

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j \quad , \quad P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$

AI Models as Evaluator

- Evaluation for open-ended generation
  - GPT-4 for evaluating MLLMs' responses in terms of hallucination

```
Please act as an impartial and objective judge and evaluate the quality
    of the response provided by a Large Multimodal Model (LMM) to the
    user question. Your evaluation should be mainly based on whether the
    response is informative, and whether the response contains any
    hallucination. Hallucination, in this context, refers to a situation
    where the LMM generates a response that includes information not
    present or implied in the image or previous conversation. A
    hallucination could be a false claim about an object, action, emotion
    , or any other detail that is not grounded in the image.

For clarity, consider these examples:

### Example 1:
- Image Contents: Lynx, Cat
- Question: How many cats can you see in this image?
- Standard Human-Generated Answer: There are two cats visible in the
    image.
- LMM Response to Evaluate: There is one cat in the image.
- Analysis: The LMM's response is false because it claims there is only
    one cat in the image, when in fact there are two.
- Hallucination: Yes.
```

AI Models as Evaluator

- Evaluation for open-ended generation
  - GPT-4 for evaluating MLLMs' responses in terms of hallucination
    - GPT-4 receives model's prediction and human annotation as well as manual annotated objects in images
    - Evaluation for hallucination in responses

AI Models as Evaluator

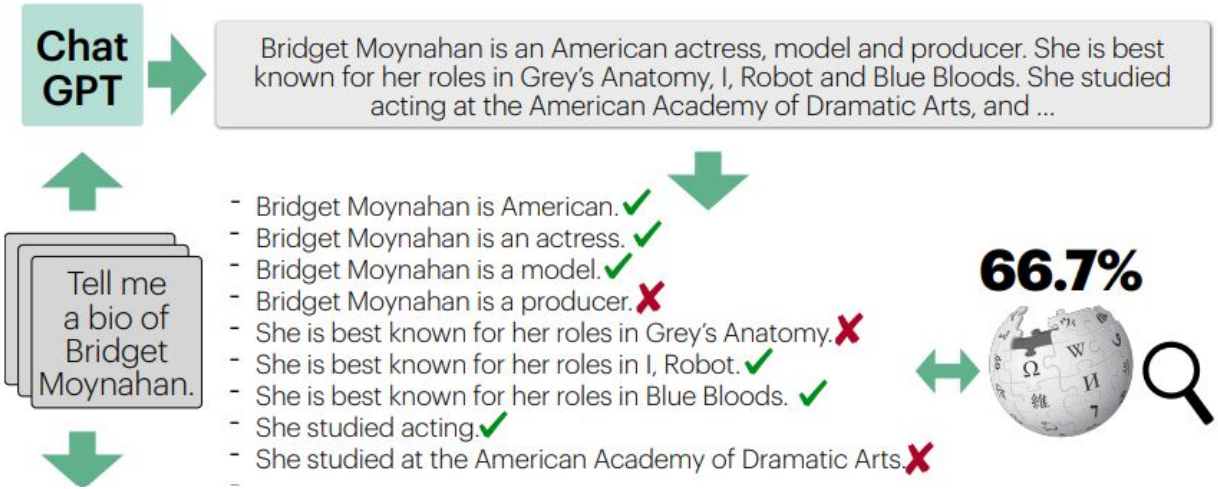
- Limitations
  - Length responses: exceed 100 words in image description
  - Compositional response:
    - Object detection, counting, spatial reasoning etc.
    - Compositional reasoning is hard

Generate Detailed Image Description

The image ... The first sign displays ... "Shakespeare," while ... shows "West Beaumont." ... In the background, there are two people standing to the left of the sign ... A car ...

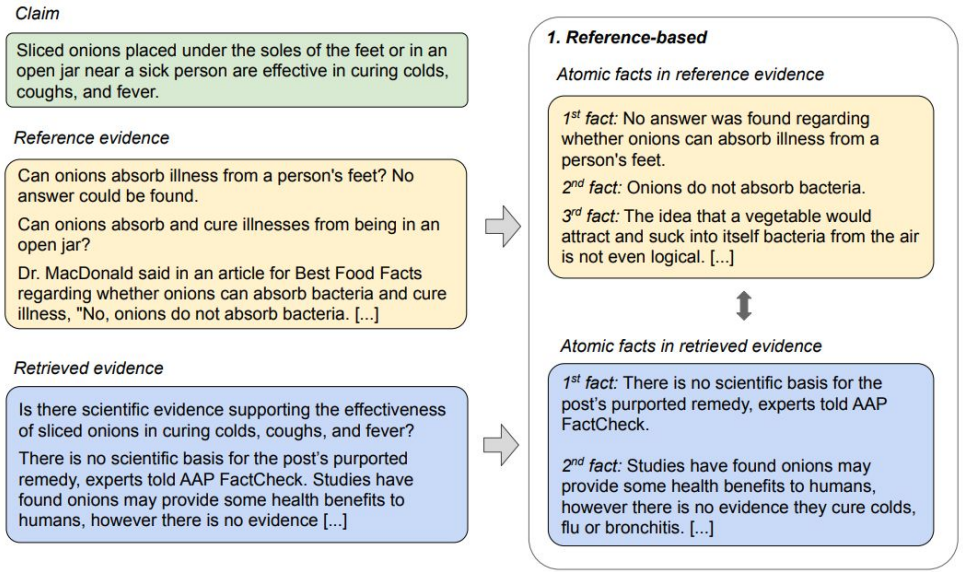
AI Models as Evaluator

- Decompose and evaluate: fine-grained and stable evaluation
  - Decompose responses into atomic facts
  - Evaluate each atomic fact



AI Models as Evaluator

- Decompose and evaluate: Ev2R
  - Evaluation of retrieved evidence from open-web



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ANY QUESTION

Safety in Agentic Pipeline

- Over trust in responses
  - Bias in AI models: certain political leaning
  - Give you a final answer, while imperceptible to retrieved evidence
    - Exacerbate the bias?

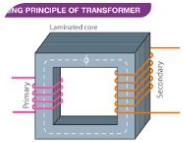
Safety in Agentic Pipeline

- Over trust in responses
  - Factuality in responses

Google

what is transformer

A transformer is **a passive electrical device that transfers electrical energy between two or more circuits through electromagnetic induction, usually to step voltage up or down while maintaining frequency**. It consists of primary and secondary wire coils (windings) around a magnetic core, facilitating efficient power distribution.

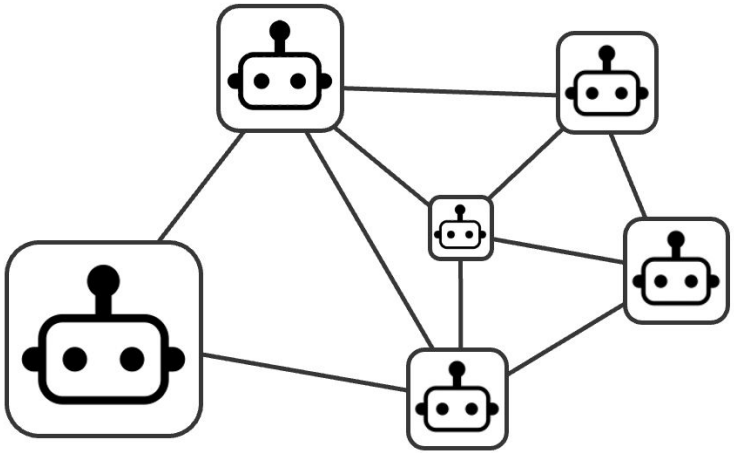


Key Aspects of Electrical Transformers:

- **Function:** They change AC voltage levels to allow safe, efficient long-distance power transmission.

Safety in Agentic Pipeline

- Error propagation in pipelines
  - Human-computer-interaction
  - Continual learning from errors



Image